A4.5 Mismatched memory attributes

Multiple agents that are accessing the same area of memory, can use mismatched memory attributes. However, for functional correctness, the following rules must be obeyed:

- All masters accessing the same area of memory must have a consistent view of the cacheability of that area of memory at any level of hierarchy. The rules to be applied are:

  Address region not Cacheable

  All masters must use transactions with both **AxCACHE[3:2]** deasserted.

  Address region Cacheable

  All masters must use transactions with either of **AxCACHE[3:2]** asserted.

- Different masters can use different allocation hints.
- If an addressed region is Normal Non-cacheable, any master can access it using a Device memory transaction.
- If an addressed region has the Bufferable attribute, any master can access it using transactions that do not permit bufferable behavior.

  Note

  For example, a transaction that requires the response from the final destination does not permit bufferable behavior.

A4.5.1 Changing memory attributes

The attributes for a particular memory region can be changed from one type to another incompatible type. For example, the attribute can be changed from Write-through Cacheable to Normal Non-cacheable. This requires a suitable process to perform the change. Typically:

1. All masters stop accessing the region.
2. A single master performs any required cache maintenance operations.
3. All masters restart accessing the memory region, using the new attributes.

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